



Abdulmalik Abdulmawla

SPATIAL-DATA TOOLS: ARCHITECTURE, URBAN AND RURAL SPACE, MOBILITY, CITY ECONOMY

PROFILE

I am interested in solving spatial problems, in architecture, urban and rural space, economics and mobility. I believe that any complex challenge, and any handling of data, requires intelligently breaking the problem down to its core parameters and simplifying each element until a humanly manageable scale is reached. This is how nature works: complex systems are not made of complex parts, but of very simple ones, intelligently put together.

I come from research. Technology is one of my strengths; my main one is computational thinking about spatial data problems, together with the capacity to work my way into the field the data serves, from simple to complex objectives. I work on the systems behind the tools, from agent-based simulation to AI-assisted, multi-agent coding, and see early where users will stumble.

EXPERIENCE

Dec 2023 – Sep 2026

Research Associate, Chair of Transport System Planning

Bauhaus-Universität Weimar · Bauhaus.Mobility / European Digital Innovation Hub Mid Germany (formerly EDIH Thuringia)

Interactive geodata platforms, among them the browser-based analysis platform for Germany's official accident data (sole developer; the official M-Uko 2012 method as an interactive tool); data-science consulting and training for public administrations in open geodata; supervision of student projects.

May 2016 – Mar 2023

Research Associate, Computer Science in Architecture (InfAR, now InfAU)

Bauhaus-Universität Weimar

Teaching computational methods for urban planning and analysis; development of design and research tools (DeCodingSpaces Toolbox, mineR); supervision of master's theses; conference papers (Space Syntax Symposium, eCAADe).

Dec 2017 – Jan 2019

Project Coordinator, DAAD exchange project "Discovering Cities"

Bauhaus-Universität Weimar and German-Jordanian University, Amman · held concurrently with the InfAR post

Project proposal, project finances, teaching seminar in Weimar, workshop in Amman.

May 2016 – 2020

Contributing Author, DeCodingSpaces Toolbox

DecodingSpaces GbR, Weimar

C# development, testing, documentation and consulting; interactive masterplan-analysis tool for Blankenburger Süden (TSPA, Berlin, 2019).

EDUCATION

Nov 2018 – present

PhD candidate, in progress · Computer Science in Architecture and Urbanism (InfAU), Bauhaus-Universität Weimar

Working title: "Spatial dependency of urban retail: detailed methods for positioning street-level activities" · supervisor: Prof. Dr. Reinhard König

2013

Master of Arts in Architecture · Hochschule Anhalt, Dessau-Roßlau, Germany

Thesis: "Heterogeneous-Atelier Industry" (with Ali Farhan): Şişhane's atelier networks, mapped and simulated as one bottom-up factory

2007

BSc Architecture and Engineering · Mansoura University, Egypt

CONTACT

LOCATION

Weimar, Germany

EMAIL

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PORTFOLIO

abdulmalikabdulmawla.github.io

GITHUB

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LANGUAGES

Arabic · native

English · fluent

German · very good

SKILLS

Spatial data and analysis

Official German geodata (ALKIS, cadastre, LoD2/CityGML, WFS/OGC), OpenStreetMap, spatial SQL (DuckDB, GeoParquet), QGIS, network analysis, space syntax, statistics (R)

Tool and application development

Python, TypeScript/JavaScript (React, deck.gl), C# (Grasshopper), simulation and agent-based models, Git, continuous integration, AI-assisted development (Claude Code)

Teaching and consulting

Seminars, workshops, training public administrations in working with open geodata

Design and visualisation

Rhino, AutoCAD, Revit, Adobe Photoshop, Illustrator, InDesign

TEACHING

- 2023 – 2026** · Digital Traffic
Simulation Lab · Open Mobility
Data · Rent-a-Data-Scientist ·
EDIH+BM.VSP.BUW
- 2016 – 2023** · Algorithmic
Architecture · Parametric Urban
Design and Analysis · SYNCITY I
and II · Circular Urbanism ·
InfAU.BUW
- 2018** · Discovering Cities teaching
seminar (DAAD) · InfAU.BUW
- 2014 – 2016** · Design 5 Studio ·
Web Application in Architecture
· AR.AASTMT
- Supervision of master's theses
and projects · 2016 – 2023
InfAU.BUW · 2023 – 2026
DTRL.VSP.BUW

WORKSHOPS (ORGANISER)

- 2018** · Discovering Cities
Workshop · GJU, Amman
- 2018** · Urban Analysis, Synthesis
and Exploration with
Grasshopper · eCAADe, Łódź
- 2016** · Rural-to-Urban
Transformation Workshop · ECL,
Addis Ababa
- 2015** · Simulating Agent-Based
Systems using Processing ·
Cairo

RECOGNITION

- 2026** · Ten-year service
recognition at the Bauhaus-
Universität Weimar, presented
by the University President

COMPETITIONS

- 2014** · Lusail Underpasses Public
Art Competition, phases 1 and
2, Doha
- 2007** · 2nd place, 4th National
Architecture Students
Competition, Cairo University
- 2006** · 1st place, 3rd National
Architecture Students
Competition, Cairo University

BUW Bauhaus-Universität Weimar · VSP
Verkehrssystemplanung · InfAU
Informatik in der Architektur und
Urbanistik · EDIH+BM European Digital
Innovation Hub Mid Germany,
Bauhaus.Mobility · DTRL Digital Transport
Research Lab · AASTMT Arab Academy
for Science, Technology and Maritime
Transport, Cairo · AR Architecture
Department · GJU German-Jordanian
University · ECL Emerging City Lab

EXPERIENCE (CONTINUED)

**Oct 2014 – Apr
2016**

Research Associate, Teaching and Computational Design

Arab Academy for Science, Technology and Maritime Transport
(AASTMT), Cairo

Teaching computational and parametric design (Grasshopper,
Processing, Arduino) in the RIBA Part I programme.

**Jan 2012 – Apr
2016**

Co-founder and Architect, Gozour Studio

Mansoura, Egypt

Residential, education and competition projects, among them the Lusail
Underpass Competition, Qatar.

**May 2009 – Sep
2011**

Junior Architect, EAI Group, Abu Dhabi

Elite Design and Engineering Consultancy, semi-government: client
representative for outsourced projects, client consultant for in-house
projects

**Jul 2008 – Apr
2009**

Freelance Architect, Egypt

3D modelling and visualisation for residential construction projects

**Jul 2007 – Jun
2008**

Junior Architect, Invert Studios, Cairo

Architectural design and planning at the office of Kamel Louqman and
Amir Wahid

Jun – Aug 2005

Intern, Al Turath Engineering Consultancy, Abu Dhabi

CAD design, layout and plotting of complete project sets

SELECTED PUBLICATIONS

- 2022** **Abdulmawla, A.**, Bielik, M., Schneider, S., König, R. Mapping active frontages: a method for linking street-level activities to their surrounding urban forms. A case study of the city of Weimar, Germany. *13th International Space Syntax Symposium*, Bergen.
- 2022** **Abdulmawla, A.**, Schneider, S., König, R., Bielik, M., Fuchkina, E. Parametric urban data structuring and spatial query: advanced data mapping and selection methods for parametric modelling environments. *40th eCAADe*.
- 2019** Bielik, M., König, R., Fuchkina, E., Schneider, S., **Abdulmawla, A.** Evolving configurational properties: simulating multiplier effects between land use and movement patterns. *12th International Space Syntax Symposium*, Beijing.
- 2018** **Abdulmawla, A.**, Schneider, S., Bielik, M., König, R. Integrated data analysis for parametric design environments: mineR, a Grasshopper plugin based on R. *36th eCAADe*.
- 2017** König, R., Bielik, M., Knecht, K., **Abdulmawla, A.**, Fuchkina, E. New methods for urban analysis and simulation with Grasshopper: using the DeCodingSpaces-Toolbox. *CAAD Futures 2017*, Istanbul.
- 2017** Schneider, S., **Abdulmawla, A.**, Donath, D. The effect of the street network on movement patterns and land use in small cities: a comparative study of three 10K cities in Thuringia, Germany. *11th International Space Syntax Symposium*, Lisbon.
- 2017** Dennemark, M., Schneider, S., König, R., **Abdulmawla, A.**, Donath, D. Towards a modular design strategy for urban masterplanning: experiences from a parametric urban design studio on emerging cities in Ethiopia. *eCAADe 2017*, Rome.
- 2016** **Abdulmawla, A.**, Farhan, A. Heterogeneous assemblages of atelier industries. *SimAUD 2016*, London (poster).

Full list: researchgate.net/profile/Abdulmalik-Abdulmawla